

ABOUT

Summary

Data engineer with 4+ years across web-scraping automation, ETL/data pipelines, and GenAI-powered applications. Started in Perl-based publishing automation, moved through large-scale e-commerce scraping and Bloomberg financial data engineering at TCS, and now designs distributed data infrastructure and LLM/VLM-powered extraction pipelines at Thurro — comfortable owning a problem end-to-end, from a raw HTML page or PDF to a production system serving real analytics.

CAREER

Experience

Data Engineer Thurro Chennai, India Designed and built a distributed job-execution platform (FastAPI + MySQL + APScheduler) that centrally schedules, dispatches, and tracks ~1,500 production web-scraping scripts across a multi-server AWS EC2 fleet, replacing a fragmented per-machine cron setup with zero code duplication. Architected a broker-free, database-polling job queue using row-level locking (SELECT ... FOR UPDATE SKIP LOCKED) with automatic crash detection and retry-with-backoff, deployed end-to-end on AWS EC2 Windows servers as auto-restarting native services (NSSM). Built and hardened a config-driven RAG ingestion pipeline processing 28+ financial/regulatory sources into a ClickHouse-backed vector corpus, including diagnosing a native heap-corruption crash and a cross-host n8n execution bug in production. Reverse-engineered a captcha-gated government vehicle-registration portal into a 5–10x faster automated data pipeline, uncovering and fixing a data-classification bug responsible for ~1.7M misclassified vehicle registrations. Built an LLM+VLM-powered extraction pipeline turning 40+ regulatory PDF disclosure forms across 30+ Indian insurance companies into structured financial data, using a two-pass text-then-vision verification flow. Dec 2025 – Present
Senior Analyst (Python Data Analyst) — Key Client: Bloomberg Tata Consultancy Services Chennai, India Developed Power BI dashboards to visualize Bloomberg financial datasets, enabling data-driven decisions for stakeholders. Automated data cleaning (Python/pandas) and ETL pipelines for Bloomberg market data feeds, improving processing speed by 30%. Collaborated with Bloomberg's analytics team to optimize data-retrieval queries, reducing report generation time by 25%. Designed a web scraping tool (Scrapy/Selenium) to collect supplementary financial data from public sources. Feb 2025 – Nov 2025
Software Engineer Lumina Datamatics Chennai, India eBay web scraping: end-to-end pipelines (Scrapy, BeautifulSoup, Selenium) against JavaScript-heavy, AJAX-based pages; cut scraping time 40% via multi-threaded crawling. Processed scraped data into MongoDB/MySQL; built validation/transformation scripts and integrated REST APIs for real-time sync. May 2024 – Feb 2025
Junior Software Engineer Lumina Datamatics Chennai, India JXpress publishing platform: automated XML/HTML transformations using Perl and regex-based text processing. Python/MySQL automation for content-classification, improving workflow efficiency 25% across 500+ daily manuscripts. May 2023 – May 2024
Software Engineering Trainee Lumina Datamatics Chennai, India Onboarded into production software engineering practices — agile delivery, code review, and on-call support. May 2022 – May 2023

Featured Projects

Distributed Scraper Execution Platform

Broker-free job orchestration coordinating ~1,500 scripts across a multi-server AWS EC2 fleet — Thurro

- Built a distributed job-execution platform (FastAPI+MySQL+APScheduler) coordinating ~1,500 scripts across a multi-server AWS EC2 fleet, zero code duplication.
- Broker-free job queue via row-level locking with automatic crash-recovery; deployed as auto-restarting Windows Services (NSSM), validated against 20 real production scripts.

Python FastAPI SQLAlchemy 2.0 APScheduler MySQL AWS EC2

AdQvest Document Chunking & Embedding Pipeline

Config-driven RAG ingestion pipeline turning 28+ regulatory sources into a searchable vector corpus — Thurro

- Built a config-driven pipeline unifying 28+ heterogeneous regulatory sources into a ClickHouse-backed RAG corpus via one YAML-configured engine.
- Redesigned serialized chunking into a concurrent, per-table-parallel architecture; diagnosed and fixed a native heap-corruption crash and a MySQL connection-leak cascade in production.

Python MySQL ClickHouse AWS S3 Docker VoyageAI n8n

VAHAN Vehicle Registration Data Pipeline

Reverse-engineered a captcha-gated govt. portal; found a bug behind 1.7M misclassified registrations — Thurro

- Reverse-engineered an undocumented JSON API behind a captcha-gated government portal, replacing browser-automation with a 5–10x faster direct-API pipeline across 20+ report configurations.
- Root-caused and fixed a data-classification bug conflating two EV fuel types, recovering ~1.7M misclassified vehicle registrations — proved via exact month-by-month reconciliation.

Python requests Playwright MySQL pandas Ollama Zyte

IRDAI Insurance Disclosure Data Extraction Pipeline

Two-pass LLM+VLM pipeline extracting structured data from 40+ regulatory PDF forms, 30+ insurers — Thurro

- Built an automated LLM+VLM pipeline extracting structured financial data from 40+ regulatory PDF forms across 30+ Indian insurers and 10+ fiscal quarters, eliminating manual data entry.
- Two-pass multimodal extraction (LLM text pass + VLM vision pass) with coordinate-based table reconstruction; custom compact output format (TOON) cutting token usage.

Python PyMuPDF Gemini API Ollama ClickHouse AWS S3

Indian Law AI Chatbot

Personal project — streaming legal Q&A assistant with a self-refreshing 20,700+ entry dataset

- Streamlit chatbot grounded in a 20,700+ entry curated Q&A dataset covering the Supreme Court and all 24 High Courts.
- Streaming Gemini responses with multi-key failover, a fully automated daily GitHub Actions refresh pipeline, and inline LLM-generated chart rendering.

Python Streamlit Gemini API SQLite Plotly GitHub Actions

Automated Document Intelligence Pipeline

Multi-engine scraping & dual-database pipeline for 229 financial/regulatory site sources

- Config-driven scraping framework covering 229 site configurations from one unified codebase, with multi-engine fallback (Selenium → Playwright → HTTP).
- Dual-database pipeline (MySQL+ClickHouse) with AWS S3 archival and LLM-based date resolution, across ~29K lines of Python.

Python Selenium Playwright MySQL ClickHouse AWS S3 Ollama

TOOLBOX

Skills & Technologies

LANGUAGES

Python Perl SQL

BACKEND & APIS

FastAPI APScheduler REST APIs SQLAlchemy 2.0

WEB SCRAPING & AUTOMATION

Selenium Playwright Scrapy BeautifulSoup requests

DATABASES

MySQL MongoDB SQLite ClickHouse

AI / GENAI

Gemini API Ollama VLMs RAG Pipelines VoyageAI

CLOUD & TOOLING

AWS S3/EC2 Docker NSSM n8n Zyte Git

CREDENTIALIALS

Certifications & Education

CERTIFICATIONS

- Salesforce Certified Administrator (SCA)
- Python — LinkedIn Learning
- Data Science / Machine Learning — LinkedIn Learning

EDUCATION

BCA — Computer Programming & Applications
Vel-Tech Ranga Sanku Arts College, Chennai
2018 – 2021